

Nuclear extraction and fractionation of chromatin-associated proteins

Risk assessment

▷ Wear gloves, safety glasses, lab coat



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This is a bench card. Full protocol available online.

Choosing a nuclear extraction and fractionation workflow

- (1.) Choose a strategy for extracting target proteins from subnuclear compartments. If unsure, start with salt fractionation after partial MNase digestion or total chromatin extraction.  

Target compartment	Accessibility	Buffer composition	Extraction strategy	Example markers
Nucleosol	High	<150 mM NaCl	Low salt extraction	Sp1, Pol II
Euchromatin	Intermediate	150–300 mM NaCl	Partial MNase digestion	H3K4me3, BRD4
Heterochromatin	Low	>400 mM NaCl	Partial MNase digestion	HP1, H3K9me3
Total chromatin	Varying	>400 mM NaCl	Full MNase digestion	
Insoluble histones	Very low	Not applicable	Acid extraction  SOP 0026	Histone H1
Nucleolar proteins	Very low	Not applicable	Sonication	NPM1

>> Preparing nuclei for extraction and optional MNase digestion

- ☐  R0143 Tris-NaCl-KCl, 250 µL
- ☐  R0145 0.5 M Ethylene glycol tetraacetic acid, 10 µL
- ☐  R0144 200 U/mL Micrococcal nuclease, 50 µL

- (1.) Start with a nuclei pellet isolated from 1.0×10^6 – 1.0×10^7 cells. Bring to 1.0×10^7 /mL in TNK buffer. Gently flick the tube to disperse remaining clumps. 

Critical: The nuclear membrane is fragile! Gentle pipetting is sufficient to disrupt most of the pellet. Clumping after resuspension indicates lysis — discard and repeat. If iso-osmotic sucrose buffer containing Tergitol-type NP-40 (CAS 9016-45-9) was used to prepare nuclei, highly mobile nuclear proteins like HMGB1 may already be lost. 

- (2.) *Optional:* Supplement the TNK buffer for chromatin digestion with calcium and MNase. Incubate with gentle mixing at 37 °C, and inactivate on ice. See table for common conditions. 

Target fraction	MNase	Additives	Treatment	Inactivation
Total chromatin	5.0 U/mL	1 mM CaCl_2	20 min at 37 °C	5 mM EGTA
Euchromatin	1.2 U/mL	1 mM CaCl_2	10 min at 37 °C	5 mM EGTA

Quality assurance: Optimize MNase concentration and time for each cell type. Ideal digestion yields mostly mononucleosomes with minor di-/tri-nucleosome signal. Excessive MNase treatment will chop all nucleosomal DNA down to 10 bp fragments. 

- (3.) Pellet the nuclei at $400 \times g$ for 10 min at 4 °C. If MNase was used, keep the supernatant for analysis. 
- (4.) Wash the pellet once with 2–5 vol TNK buffer and proceed to extraction.

>> Nuclear extraction and fractionation

- ☐  R0146 Nuclear extraction buffer, low-salt (LS), 10 mL
- ☐  R0147 Nuclear extraction buffer, high-salt (HS), 10 mL

- (1.) Prepare buffers with defined NaCl concentrations by mixing LS and HS nuclear extraction buffers:

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Nuclear extraction buffer	0 mM NaCl	75 mM NaCl	150 mM NaCl	300 mM NaCl	400 mM NaCl
Low-salt	1.5 mL	0.2 mL	0.4 mL	0.8 mL	1.0 mL
High-salt	0.0 mL	1.4 mL	1.2 mL	0.8 mL	0.5 mL

- (2.) Resuspend the nuclear pellet in 100 μ L extraction buffer per 1.0×10^7 nuclei. Add fresh inhibitors.

Critical: Use chilled buffers. Add protease inhibitors just before use to prevent degradation.

- (3.) Incubate on ice for 30 min. For volumes over 100 μ L, rotate at 4 °C.
- (4.) Separate soluble fraction by centrifugation at $400 \times g$ for 10 min at 4 °C. Collect the supernatant.
- (5.) Repeat the extraction using buffers with increasing salt concentration on the remaining pellet. Analyze and pool fractions as needed.
- (6.) The remaining chromatin pellet can be collected at $16\,000 \times g$ for 40 min at 4 °C and resuspended for further processing such as acidic extraction or sonication.



⌚ 30 min



🔗 Recipe (available online) 🔗 Resources (available online) 📖 Notes (available online)

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